

Annotation-Efficient Semantic Segmentation with Shape Prior Knowledge

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ABSTRACT

Deep learning methods have achieved great success on semantic segmentation in recent years. But the training typically relies on large-scale fully-annotated ground truth masks, which are difficult to obtain in practice. In this research, we study the problem of reducing the annotation cost of segmentation network training with a focus on exploring the shape prior knowledge of objects. Under the context of three applications, we study three types of shape priors. Specifically, we first exploit the implicit shape prior of curve structures to propose a weakly supervised curve structure segmentation method, and then explicitly formulate the shape prior of anatomical structures as loss functions to propose a one-shot anatomical structures segmentation network. Last, we try to generalize the shape constraint to arbitrary objects to propose a class-agnostic few-shot segmentation framework. Experiment results show that our methods could achieve comparable or better performance than fully supervised segmentation methods with less annotation costs on the studied applications.

CCS CONCEPTS

• Computing methodologies → Image segmentation.

KEYWORDS

semantic segmentation; neural networks; shape prior knowledge

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1 INTRODUCTION

As a fundamental computer vision task, image semantic segmentation has witnessed great success of deep learning models in recent years [4–6, 14, 20, 21]. However, training parameter-intensive deep learning models needs abundant fully-annotated data with ground truth labels [19]. Annotating pixel-wise ground truth segmentation masks is extremely laborious, and sometimes requires specialized

domain expertise, e.g., medical image segmentation. As a consequence, the application of deep learning models is usually limited by the lack of training data in practice.

In this research, we try to reduce the annotation cost of segmentation network training by incorporating shape prior knowledge of objects. Combining shape prior and image segmentation has been proven effective in many non-learning-based methods, such as active contour model [8], level-set [23], graph cut [17], and so on. In the age of deep learning, it has not been adequately explored, because the fully convolutional network (FCN) [9] and its variants [4, 14, 21] solve image segmentation in a manner of pixel-wise classification, which is not convenient for integrating shape-related information. To solve this issue, we propose three methods to incorporate three different types of shape priors into segmentation model training, respectively. Each method is proposed for a practical segmentation task, corresponding to curve structure segmentation, anatomical structure segmentation, and cross-domain few-shot segmentation, respectively.

In the first work, we explore the application of implicit shape prior in weak curve structure segmentation of cultural heritage objects. Due to the blurry boundary and subtle depth change, curve structures stamped on the surface of cultural heritage objects are very hard to recognize for both automatic segmentation and manually annotation. Taking advantage of the prior knowledge of their thin and elongated structure, we propose a CNN framework to extract curve skeletons instead of full curves. In this way, we simplify the annotation target from pixel-wise curves to single-pixel skeletons, thus implicitly realize weakly supervised segmentation. This work has been published in 2018 [12].

In the second work, we attempt to directly incorporate explicit shape priors into CNN loss functions to enable one-shot anatomical structure segmentation. Owing to the inherent regularized nature of anatomical structures, the same anatomy in different images share common features in shape, appearance and gradients. Human annotators are able to annotate new images by referring to a few example. Inspired by the human learning behavior, we propose to use the shared features in an exemplar segmentation to partially supervise other unlabeled images in CNN training, thus make it trainable in a one-shot setting. Experiment results demonstrate that our method performs competitively to the state-of-the-art fully supervised methods. This work has been published in 2020 [11, 22].

The last work is still in the process of implementation, where we want to extend the utilization of prior knowledge from specific objects to arbitrary objects. The problem setting is to segment any class in the query image by referring to a few support images. We first follow the idea of meta-learning in few-shot segmentation to train a base model on existing labeled datasets. To overcome the

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domain shift, we propose a semi-supervised fine-tuning strategy to fine-tune the base model with labeled support images and unlabeled query images of the target domain. It could adapt the base model to the target domain with very few labeled images.

2 CURVE STRUCTURE SEGMENTATION WITH IMPLICIT SHAPE PRIOR

Extracting curve structures from the depth images of pottery sherds is an important problem in archaeology. After thousands of years of erosion, the curve structures become extremely weak and noisy, and are difficult to segment for existing methods. Traditional non-learning-based segmentation methods are not able to handle various noises in the depth images. Moreover, the blurry boundary of curves make them difficult to annotate even for experts. Consequently, fully supervised learning-based methods are not applicable because of the lack of ground truth masks. In this work, we take advantage of the shape prior knowledge of curve structures and propose a weakly supervised CNN-based curve structure segmentation algorithm, which takes the curve skeleton map as input of training and could predict the full curve structure.

As shown in the first column of Fig. 1, the underlying designs of the curve structures bear certain geometries and patterns. For example, most of the curve structures consist of elongated smooth curve segments. Furthermore, many curves in the structures show good parallelism against each other. With this knowledge, we disentangle the curve structure into curve skeleton and curve width, because the curve skeletons are relatively easy to annotate compared to the blurry curve boundaries. The core idea is to utilize the receptive field size of different layers in FCN to simultaneously extract curve skeleton and estimate curve width. In this way, we implicitly leverage the shape prior knowledge of curve structures to reduce the annotation cost of segmentation network training.

Our method mainly consists of three steps, skeleton extraction, skeleton refining, and curve recovering. In the first step, a FCN is employed to extract the skeleton of curve structures, and estimate a scale value at each skeleton pixel. Our network is adopted from the network architecture of [10], which is composed of three pairs of cascaded convolutional encoders and decoders. To make full use of all the extracted features, the decoders are organized in a way of stepwise accumulation when fusing them together to predict the skeleton map. Since different encoders correspond to different receptive field sizes, at each pixel the receptive field size of the encoder with the largest score reflects the scale at this pixel. In the second step, we employ a pixel classification network to refine the predicted skeleton map. The skeleton maps obtained in the first step may still contain false positives. To further improve the precision, we crop a patch around each detected skeleton pixel, and use a ConvNet to predict a binary label that indicates if it is a true skeleton pixel. Last, an adaptive thresholding algorithm is proposed to recover the refined skeleton map to full curve structures with the help of estimated curve width.

To evaluate the proposed method, we annotated the ground truth masks of 530 depth images, in which 250 images are for training and 280 images are for testing. We compare our method with both non-learning-based methods and popular CNN-based methods. The compared CNN-based models are trained with full ground truth

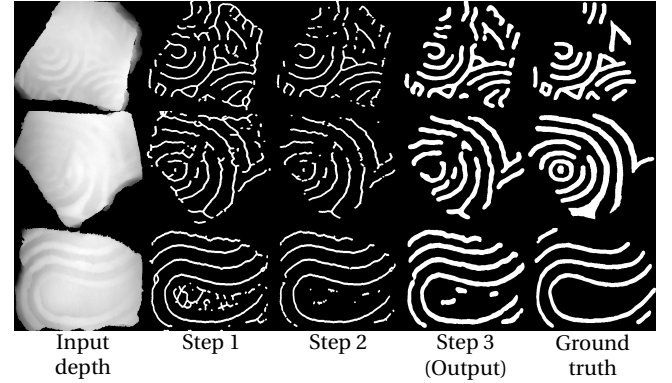


Figure 1: Example results of the proposed curve structure segmentation method.

Table 1: Precision, recall and F-score of curve structure segmentation results of 280 test images.

Methods	Precision	Recall	F-score
DoG	0.366	0.774	0.490
LevelSet [23]	0.262	0.938	0.399
GrabCut [15]	0.357	0.671	0.448
FCN [10]	0.589	0.472	0.514
DeepLab [4]	0.585	0.670	0.583
DeepSkeleton [16]	0.634	0.690	0.654
Ours	0.660	0.827	0.731

masks, while our method uses only skeleton maps. The quantitative results are reported in Table 1. Our method outperforms non-learning-based methods by a large margin, and surpasses state-of-the-art learning-based methods which are trained with full curve masks. We also show several qualitative results of our method in Fig.1, as well as the intermediate results of each step.

3 ANATOMICAL STRUCTURE SEGMENTATION WITH EXPLICIT SHAPE PRIOR

In the second work, we study the problem of anatomical structure segmentation by explicitly incorporating object prior knowledge into neural network training. Segmentation of anatomical structures serves as a core element in a wide spectrum of medical image analysis applications. Recent advances in deep learning research have significantly boosted the accuracy of medical image segmentation. However, without abundant pixel-level labels, the state-of-the-art segmentation methods [4–6, 14, 20, 21] cannot achieve their optimal performance [19].

We propose an annotation-efficient anatomical structure segmentation method, termed Contour Transformer Network (CTN). Our work is inspired by the human annotator’s capability of learning segmentation of anatomical structure from one or very few exemplars. This is achieved by understanding the shape and appearance traits of the target object from the exemplars and actively

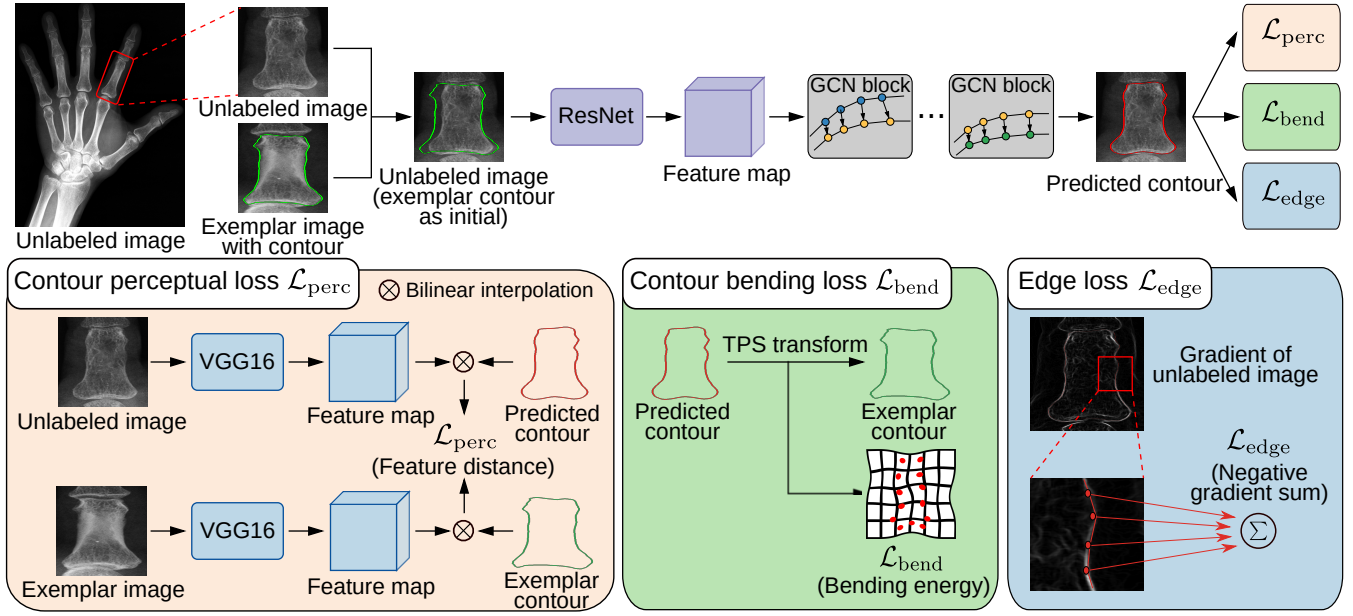


Figure 2: Contour Transformer Network (CTN) proposed for one-shot anatomical structure segmentation.

looking for objects with similar traits in new images. To mimic this behavior, we propose a semi-supervised learning approach that exploits the shape and appearance similarities of the target object between labeled and unlabeled images to train a segmentation model. As a result, CTN is able to learn segmentation from one labeled exemplar and a set of unlabeled images without dependency on external labeled datasets.

Owing to the inherent regularized nature of anatomical structures, the same anatomy in different X-ray images may share common features or properties, such as the anatomical structure’s shape, appearance and gradients along the structural object boundary. Although different images are not directly comparable, we can compare their common features only and use the exemplar segmentation to guide other unlabeled images partially, thus making CTN trainable in a one-shot setting. Specifically, we formulate the segmentation problem as learning a contour evolution behavior modeled by a cascaded graph convolutional network (GCN). Three differentiable contour-based loss functions namely contour perceptual loss, contour bending loss and edge loss are proposed to describe the common features of appearance, shape and edge response, respectively. For each unlabeled image, CTN takes the exemplar contour as an initialization, then gradually evolves it under the guidance from the three losses. The pipeline of CTN is illustrated in Fig. 2.

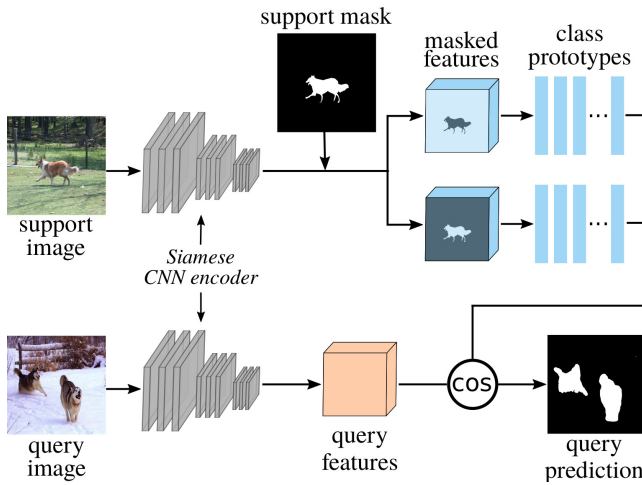
The contour perceptual loss $\mathcal{L}_{perc}$ is partially enlightened by the perceptual loss [7] developed for image super-resolution, which measures image perceptual similarities in the feature space of VGG-Net [18], we measure contour perceptual similarities in the graph feature space. In particular, graph features are extracted from the VGG-16 feature maps of the two images along the two contours, and their L1 distance is calculated as $\mathcal{L}_{perc}$. The contour bending loss $\mathcal{L}_{bend}$ is proposed based on the prior knowledge that the same

anatomical structure on different images should have similar shape. It is calculated as the bending energy of the TPS warping [1] that maps the exemplar contour to the predicted contour. Since the bending energy measures the magnitude of the 2nd order derivatives of the warping, $\mathcal{L}_{bend}$ penalizes more on local and acute shape changes, which are often associated with mis-segmentation. Although the contour perceptual and bending losses can achieve robust segmentation, they are inherently insensitive to small segmentation fluctuations, such as deviations from the correct boundary by a few pixels. In order to obtain desirably high segmentation accuracy, we last employ an edge loss $\mathcal{L}_{edge}$ that measures the image gradient magnitude along the contour, which attracts the contour toward edges in the image.

In the evaluation, we compare our method with two non-learning-based methods MorphACWE [2, 13] and MorphGAC [3, 13], two one-shot segmentation methods CANet [24] and Brainstorm [25], and three fully supervised methods UNet [14], DeepLab-v3+ [4] and HRNet-W18 [21]. The experiments are conducted on four datasets of anatomical structures, knee, lung, phalanx, and hip. The mean IoU and Hausdorff distance (HD) of all methods are reported in Table 2. We can see that 1) CTN significantly outperforms traditional non-learning-based methods by at least 15.63% in terms of IoU; 2) Compared with other one-shot learning methods, CTN takes the prior knowledge into consideration and achieves at least 13.92% higher IoU; 3) CTN performs competitively to the state-of-the-art fully supervised methods, while using much less ground truth labels.

Table 2: The mean IoU and Hausdorff distance (HD) of CTN and seven existing methods on four datasets.

Methods		Knee		Lung		Phalanx		Hip		Mean	
		IoU(%)	HD(px)	IoU(%)	HD(px)	IoU(%)	HD(px)	IoU(%)	HD(px)	IoU(%)	HD(px)
Non-learning-based	MorphACWE [2, 13]	65.89	54.07	76.09	55.35	74.33	69.13	48.05	94.11	66.09	68.17
	MorphGAC [3, 13]	87.42	15.78	70.79	45.67	82.15	24.73	83.42	32.20	80.95	29.60
One-shot	CANet [24]	29.22	175.86	56.90	73.46	60.90	67.13	48.89	88.39	48.98	101.21
	Brainstorm [25]	90.17	29.07	77.13	43.28	80.05	30.30	82.48	44.17	82.46	36.71
	CTN (Ours)	97.32	6.01	94.75	12.16	96.96	8.19	97.29	8.27	96.58	8.66
Fully supervised	UNet [14]	96.60	7.14	95.38	12.48	95.76	10.10	96.51	13.28	96.06	10.75
	DeepLab [4]	97.18	5.41	96.18	10.81	97.63	6.52	97.64	6.24	97.16	7.25
	HRNet [21]	96.99	5.18	95.99	10.44	97.47	7.03	97.66	7.57	97.03	7.56

**Figure 3: The architecture of prototypical network for few-shot segmentation.**

4 ARBITRARY OBJECT SEGMENTATION WITH CLASS-AGNOSTIC PRIOR

In the third work, we try to generalize the utilization of shape prior knowledge in segmentation network to arbitrary objects. This work is still in progress. Our goal is to develop a method that could learn to segment any unseen objects from very few labeled examples, which is also known as few-shot segmentation. Meta-learning has been a dominant paradigm adopted by most few segmentation methods. Instead of training class-specific segmentation models, they leverage existing labels of some base classes to train a class-agnostic model which could expediently learn to segment novel classes from a few support images in test. A common problem of these methods is that they consider only in-domain few-shot segmentation, which mean the base classes and novel classes always come from the same domain. When there exists a significant domain shift (e.g., from natural images to medical images), these meta-learning models will fail because of the different feature distribution.

Prototypical network is a common meta-learning-based few-shot segmentation method, as depicted by Fig. 3. It generates class-wise

prototypes from support images, and make predictions on query images by finding the closest prototype for each pixel [20]. Pixels from the same semantic class are expected to be closer than pixels from different semantic classes in the embedding space. However, when testing on unseen classes, the different data distribution will make the embedding space less discriminative, and the less separable features usually result in false predictions.

Training a universal model on base classes that works on all classes may not be realistic. Considering the potential domain shift, we think it is necessary to adapt the base model to the target domain using available labeled and unlabeled images before testing. Our preliminary idea is to fine-tune the prototypical model with a few labeled images and a set of unlabeled images of the testing class. Specifically, we first obtain a base model by training the prototypical network on base classes. For each class, we take prototypes generated from the predicted query mask as anchor features, support prototypes from the same class as positive features, and support prototypes from all other classes as negative features. By minimizing the distance between anchor and positive features and maximizing the distance between anchor and negative features with a triplet loss, we optimize the segmentation of query images and improve the discriminability of the base model on the test class. This is a transductive fine-tuning process, but the fine-tuned model can be used to predict new query images of the test class without further ado, because the prototypical model has been adapted to the target domain. The experiment on this idea is still in progress.

5 CONCLUSION

We study the problem of annotation-efficient semantic segmentation with a focus on utilizing shape prior knowledge. Three types of shape priors are discussed, including implicit, explicit, and class-agnostic shape priors, corresponding to three applications, respectively. We first propose a weakly supervised segmentation method for curve structures segmentation, which reduces the annotation cost from pixel-wise mask to skeleton map. Next, we try to train an anatomical structure segmentation network with only one labeled image, where the object shape prior is explicitly incorporated into network training in the form of an exemplar contour. Last, we want to build a cross-domain few-shot segmentation framework that could learn to segment any novel objects from very few examples. We plan to use a semi-supervised pretraining and fine-tuning pipeline to solve this problem.

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